

Final Term Project Proposal

Instructor: Saim Sanan

Diabetes Onset Prediction Using Machine Learning

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Project Overview

Project Title	Diabetes Onset Prediction Using Machine Learning
Dataset Name	Pima Indians Diabetes Database
Dataset Source	Kaggle / UCI Machine Learning Repository
Dataset Link	kaggle.com/datasets/uciml/pima-indians-diabetes-database
Target Variable	Outcome (0 = Non-Diabetic, 1 = Diabetic)
Task Type	Binary Classification
Group Size	2 Members

Problem Statement

Diabetes is one of the most prevalent chronic diseases worldwide, affecting millions of people and leading to serious health complications if not detected early. Early diagnosis is critical for effective treatment and prevention of complications such as heart disease, kidney failure, and vision loss.

This project aims to build a machine learning classification system that can predict whether a patient is likely to develop diabetes based on key medical diagnostic features such as glucose level, BMI, age, insulin level, and blood pressure. The system will assist in early risk identification and support clinical decision-making.

Dataset Description

The Pima Indians Diabetes Database, originally from the National Institute of Diabetes and Digestive and Kidney Diseases, contains diagnostic measurements for 768 female patients. The

dataset has the following features:

Feature	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration (2-hour oral glucose tolerance test)
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-Hour serum insulin (mu U/ml)
BMI	Body mass index (weight in kg / height in m squared)
DiabetesPedigreeFunction	Likelihood of diabetes based on family history
Age	Age of the patient (years)
Outcome	Target variable: 1 = Diabetic, 0 = Non-Diabetic

Target Prediction / Objective

The primary objective of this project is to predict the onset of diabetes in patients based on their diagnostic measurements. Specific objectives include:

- Predict whether a patient is diabetic (1) or non-diabetic (0) — binary classification • Identify the most significant features contributing to diabetes risk
- Compare the performance of at least two machine learning models
- Handle missing/zero values and apply proper data preprocessing techniques • Evaluate models using appropriate classification metrics
- Visualize data distributions, feature correlations, and model results

Proposed Machine Learning Models

The following two models will be implemented and compared:

Model	Why Selected	Key Advantage
Logistic Regression	Strong baseline for binary classification; interpretable coefficients	Easy to explain feature weights during viva
Random Forest	Handles non-linear patterns; robust to outliers and missing values	Provides feature importance scores for analysis

Machine Learning Workflow

1. Data Collection & Loading

- Load Pima Indians Diabetes dataset (CSV format, 768 rows x 9 columns) •
- Inspect data types, shape, and initial statistics

2. Data Preprocessing

- Identify and handle zero values in Glucose, BloodPressure, BMI, Insulin, SkinThickness (biologically invalid zeros)
- Replace zeros with median/mean values for respective columns
- Check for duplicate records and missing values
- Normalize/standardize features using StandardScaler

3. Feature Engineering & Selection

- Analyze feature correlation using a heatmap
- Select most relevant features based on correlation analysis and feature importance •
- Apply train-test split (80% train / 20% test)

4. Model Training

- Train Logistic Regression with default and tuned hyperparameters
- Train Random Forest Classifier with n_estimators tuning

5. Model Evaluation

- Accuracy, Precision, Recall, F1-Score
- Confusion Matrix for both models
- ROC-AUC Curve comparison

6. Visualization

- Distribution plots for each feature
- Correlation heatmap
- Confusion matrix heatmaps
- ROC curve comparison chart
- Feature importance bar chart (Random Forest)

Tools & Technologies

Category	Tool / Library
Programming Language	Python 3.x
IDE / Environment	Jupyter Notebook / Google Colab
Data Manipulation	Pandas, NumPy
Machine Learning	Scikit-learn (LogisticRegression, RandomForestClassifier, StandardScaler, train_test_split, metrics)

Data Visualization	Matplotlib, Seaborn
Dataset Source	Kaggle (UCI Pima Indians Diabetes Database)
Version Control	GitHub (optional)

Deliverables

- Jupyter Notebook with complete source code
- Dataset file (CSV) or dataset link
- Presentation Slides (.pptx)
- Final Report (short written summary)